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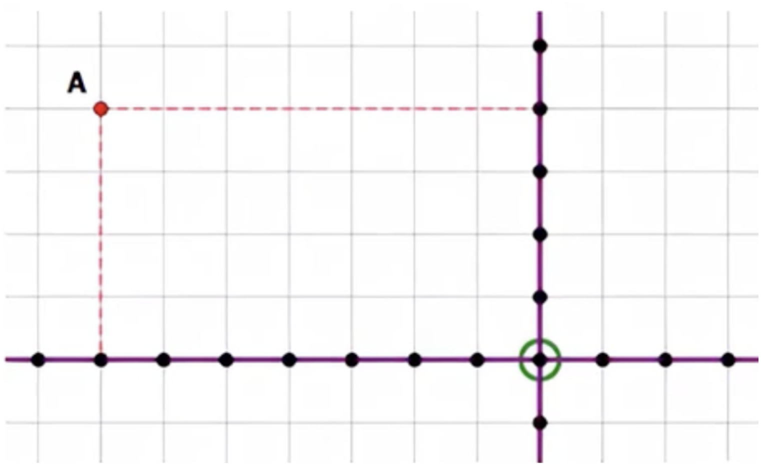
GRE Coordinate Geometry

1 Intro to Data Interpretation

1.1 The Coordinate Plane

1.1.1 Example

What are the coordinates of point A?



The axes divide the entire plane into four regions known as quadrants.

These quadrants are denoted counterclockwise from the upper right, as I, II, III, and IV.

1. In Q1, both $x > 0$ and $y > 0$
2. In Q2, $x < 0$ but $y > 0$
3. In Q3, both $x < 0$ and $y < 0$
4. In Q4, $x > 0$ but $y < 0$

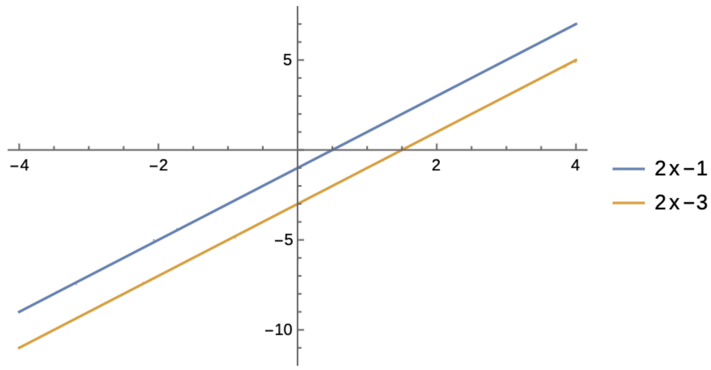
Note that any point on the x-axis or y-axis is not in any of the quadrants.

1.1.2 Problem

Point M is the midpoint of segment AB. If $A = (2, -3)$ and M is on the negative x-axis, in what quadrant is B?

1.2 Graphing lines

Every possible line in the Cartesian Plane has its own unique equation.



Technically, two points are enough to determine a line.

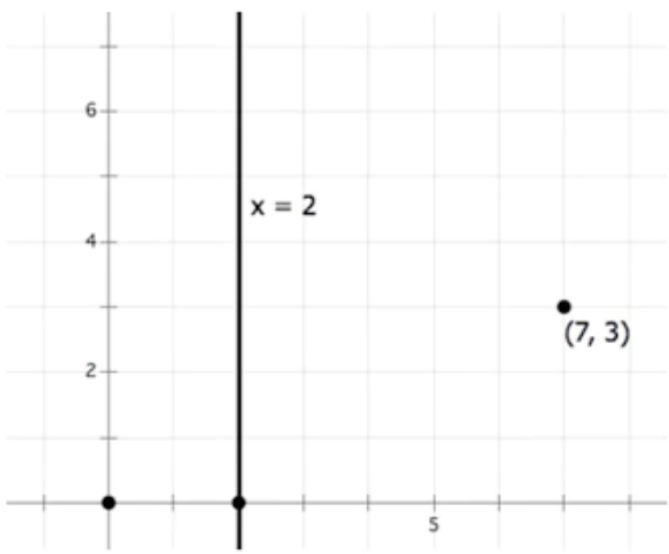
Any point on the line has to satisfy the equation of the line.

The horizontal lines have the form $y = c$, where $c \in \mathbb{R}$ and the vertical lines have the form $x = c$, where $c \in \mathbb{R}$.

If point C has the same x-coordinate as point A and the same y-coordinate as point B, then the angle $ACB = 90°$

1.2.1 Problem

In the coordinate plane, the line $x = 2$ and the points $(7, 3)$ are shown. What is the equation of a horizontal line above the point $(7, 3)$ that is as far from the point $(7, 3)$ as is the line $x = 2$



1. $x = 5$
2. $y = 5$
3. $y = 8$
4. $x = 12$

5. $y = 12$

1.3 Slope

Slope is, roughly, a measure of how steep a line is.

One of the most common definitions of slope is "rise over run". What does it mean?

Suppose we want to find the slope between two points.

The run is the horizontal separation from left to right. This is always positive from left to right.

The rise is the vertical separation between the two points as we go from left to right. If the point on the right is higher, then the rise is positive; if lower, negative.

Suppose we have two general points (x_1, y_1) and (x_2, y_2)

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$$

1.3.1 Problem

If a line goes through $(2, -1)$ and has a slope of $m = \frac{5}{3}$, find all the points (a, b) on the line where a and b are integers whose absolute values are less than or equal to 10.

First, move right: that is, right 3, up 5. $(5, 4)$, $(8, 9)$

Now, move left: that is, left 3, down 5. $(-1, -6)$

1.3.2 Properties:

- Parallel lines have equal slopes.
- Vertical lines have $\lambda_1 \cdot \lambda_2 = -1$. So they are opposite-signed reciprocals.

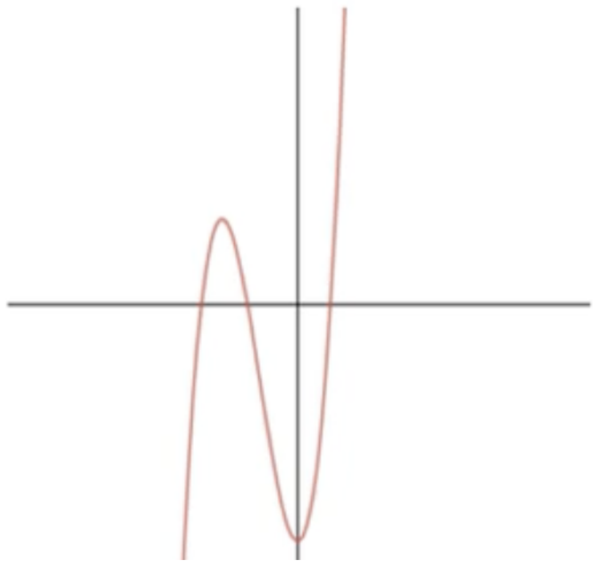
1.4 Intercepts

An intercept is the point at which a graph crosses the x- or y-axis.

- At a y-intercept, the value of $x = 0$.
- At an x-intercept, the value of $y = 0$.

1.4.1 Problem

Which equation could be represented by the graph below?



1. $y = x^3 + 4x^2 - 7$
2. $x^3 + 2s^2 - x + 7$
3. $(x^2 - 3)(x - 7)$
4. $y = x^5 + 3$

1.5 Slope-Intercept Form

If we solve the equation of a line for y, getting y on one side by itself equal to everything else, then we automatically put the equation into slope-intercept form.

1.5.1 Problem

Find all the points (j, k) on the line $y = -\frac{3}{4}x + 2$ such that j and k are both integers with absolute values less than or equal to 10.

The slope $\lambda = -\frac{3}{4}$ and the y-intercept is $(0, 2)$

Move right 3, down 4. $(3, -2), (6, -6), (9, -10)$

move left 3, up 4 $.(-3, 6), (-6, 10)$

So there are 6 possible points.

Notice that vertical lines have undefined slopes.

1.6 Equation of the Line

- The slope-intercept form, $y = mx + b$, is a fundamental tool for constructing line equations when the slope and a point on the line are known.

- Solving for the y-intercept (b) involves substituting known values into the slope-intercept equation.
- Two points uniquely determine a line, and knowing how to find the slope and y-intercept from these points allows for the construction of the line's equation.

1.6.1 Problems

1. Line A has a slope of $-\frac{5}{3}$ and passes through the point (-2.7). What is the x-intercept of line A?
2. Line J passes through the points $(-3,-2),(1,1)$, and $(7,Q)$. Find the value of Q.

1.7 Distance Between Two Points

If two points are on the same horizontal line, all we have to do is subtract the x-coordinates. Distance is always positive.

Similarly, if two points are on the same vertical line, we subtract the y-coordinates.

Let's think about two points diagonally separated:

Let x_A, y_A and $B(x_B, y_B)$ so the distance between these two points is given by

$$d = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$$

1.7.1 Problem

1. Find the distance between $(-5,-1)$ and $(3,3)$.
2. A circle in the x-y plane has a center of $(6,3)$ and a radius of 5. Find the two x-intercepts.

1.8 Reflections in the x-y Plane

- Reflections over the x-axis and y-axis involve maintaining the same x or y-coordinate, respectively, while inverting the sign of the other coordinate.
- Reflecting a point over the line $y = x$ results in the x and y coordinates being switched, while reflecting over the line $y = -x$ switches the coordinates and inverts their signs.

1.8.1 Problem

Points A(5,2) and B(-2,-5) are two vertices of an isosceles triangle. If C is the third vertex and has a y-coordinate of 4, what is the x-coordinate of C?

1.9 Graphs of Quadratics

The graph of a quadratic equation in the x-y plane is a parabola.

- Parabolas can open upwards or downwards and vary in width; their shape is determined by the coefficient of x^2 in the equation.

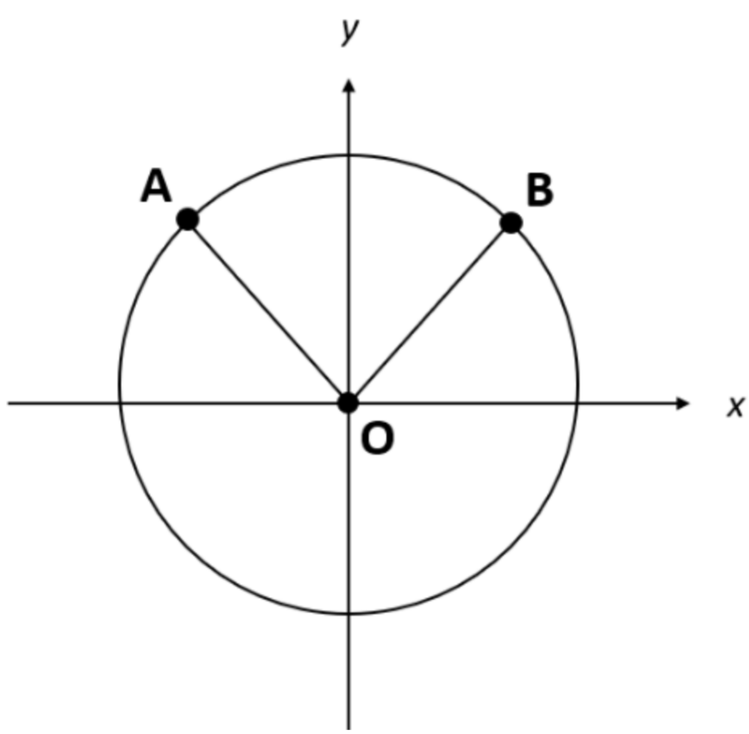
- Every parabola has a vertical line of symmetry and a vertex, which is the highest or lowest point depending on the direction the parabola opens.
- The standard form of a quadratic equation is $y = ax^2 + bx + c$, with the sign of 'a' indicating the direction of opening and its absolute value indicating the width of the parabola.
- X-intercepts of the parabola are the solutions to the quadratic equation set to zero, which can be none, one, or two, reflecting points where the parabola crosses the x-axis.

1.9.1 Problem

- A parabola with a vertex (2,1) has a y-intercept of 9. What is the x-coordinate of the other point on the parabola with a y-intercept of 9?
- A parabola has an x-intercept at $(-4, 0)$. If the vertex is at $(2, 5)$ find the other x-intercept.

1.9.2 Questions

1. In the xy-plane, a quadrilateral has vertices at $(-5, 3), (3, 3), (-5, -4)$, and $(3, -4)$. What is the perimeter of the quadrilateral?
2. What are the x-intercepts of the parabola defined by the equation $y = 2x^2 - 8x - 90$?
3. In the standard x-y plane, what is the slope of the line that is perpendicular to the line $12x + 3y = 7$?



4. Point O is the origin and the center of the circle. If the circumference of the circle is 16 π, and the arc length AB is 4π, what is the length of the straight line, not shown, connecting points A and B?